

Calculus Mentor Dialogue

4. The Dialogue

Lesson 1: Limits

Mentor: First video: Khan Academy, "Introduction to limits":

<https://www.youtube.com/watch?v=riXcZT2ICjA>

Short summary: a limit asks what value a function approaches near a point. The function may be undefined at the point itself, but the limit can still exist if nearby values approach one number. A common case is a rational expression with a removable hole.

Practice task:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Show the factoring, cancellation, final limit, and the hole.

Student: I factor $x^2 - 4 = (x - 2)(x + 2)$. Then I cancel $x - 2$, so the answer is $x + 2$. The hole is at $x = 2$.

Mentor: You did the factoring correctly, but you stopped one step early. $x + 2$ is the simplified expression, not the final limit. What value does $x + 2$ approach as $x \rightarrow 2$?

Student: It approaches $2 + 2 = 4$. So the limit is 4, and the hole is at $(2, 4)$.

Mentor: Correct. Transfer check: what is

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}?$$

Student: $x^2 - 9 = (x - 3)(x + 3)$, so after canceling I get $x + 3$. At $x = 3$, the limit is 6. The hole is at $(3, 6)$.

Mentor: Passed. You can now distinguish the simplified expression from the actual limit.

Lesson 2: One-Sided Limits and Continuity

Mentor: Video: Khan Academy, "One-sided limits from graphs":

<https://www.youtube.com/watch?v=nOnd3SiYZqM>

Short summary: a two-sided limit exists only if the left-hand and right-hand limits match. A function is continuous at a point only if the limit exists, the function value exists, and both are equal.

Practice task:

$$g(x) = \begin{cases} x + 1, & x < 3 \\ 7, & x \geq 3 \end{cases}$$

Find the left limit, right limit, function value, and continuity at $x = 3$.

Student: Left limit is 4. Right limit is also 4, because x is still approaching 3. Function value is 7. I think maybe it is continuous because $g(3)$ exists.

Mentor: This is the rule-choice mistake. From the right side, the formula is not $x + 1$; it is the constant 7. Also, having $g(3)$ exist is not enough for continuity. Try again.

Student: Left limit: $3 + 1 = 4$. Right limit: 7. Function value: $g(3) = 7$. Since $4 \neq 7$, the two-sided limit does not exist, so the function is not continuous.

Mentor: Correct. Transfer check: if the second piece were 4 instead of 7, would it be continuous?

Student: Yes. Left limit would be 4, right limit would be 4, and $g(3) = 4$.

Mentor: Passed.

Lesson 3: Derivative as Rate of Change

Mentor: Video: Khan Academy, "Derivatives introduction": <https://www.youtube.com/watch?v=N2PpRnFqnqY>

Short summary: a derivative is instantaneous rate of change. It is the limit of average rates of change:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Practice task: compute $f'(3)$ for $f(x) = x^2$ using the definition.

Student: $f(3+h) = (3+h)^2 = 9 + 6h + h^2$, and $f(3) = 9$. So

$$\frac{f(3+h) - f(3)}{h} = \frac{6h + h^2}{h} = 6 + h$$

As $h \rightarrow 0$, $f'(3) = 6$.

Mentor: Good. What does 6 mean if $f(x)$ is position in meters and x is time in seconds?

Student: At 3 seconds, the position is changing at 6 meters per second.

Mentor: Transfer check: what would $f'(a)$ be for $f(x) = x^2$?

Student: $(a+h)^2 - a^2 = 2ah + h^2$. Divide by h : $2a + h$. Limit is $2a$.

Mentor: Passed. You can connect the formula to rate of change.

Lesson 4: Power Rule

Mentor: Video: Khan Academy, "Taking derivatives": <https://www.youtube.com/watch?v=rAof9Ld5sOg>

Short summary: the power rule says

$$\frac{d}{dx}x^n = nx^{n-1}$$

Constant multiples stay in front, and constants have derivative 0.

Practice task: differentiate $y = 5x^4 - 3x^2 + 8$, then find the slope at $x = -1$.

Student: $y' = 20x^3 - 6x + 8$. At $x = -1$, that gives $-20 + 6 + 8 = -6$.

Mentor: You made the common constant mistake. The derivative of 8 is 0, not 8. Correct the derivative and the slope.

Student: Right. The derivative is

$$y' = 20x^3 - 6x$$

At $x = -1$:

$$20(-1)^3 - 6(-1) = -20 + 6 = -14$$

Mentor: Correct. Transfer check: differentiate $7\sqrt{x}$.

Student: Rewrite as $7x^{1/2}$. Then

$$\frac{7}{2}x^{-1/2} = \frac{7}{2\sqrt{x}}$$

Mentor: Passed.

Lesson 5: Product and Quotient Rules

Mentor: Video: Khan Academy, "Product rule proof": <https://www.youtube.com/watch?v=L5ErIC0COxI>

Short summary: use product rule for multiplication and quotient rule for division:

$$(uv)' = u'v + uv'$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

Practice task:

$$y = \frac{2x + 5}{x^2 + 1}$$

Set up y' using the quotient rule.

Student: I practiced this after the video. It was fine. I just used the quotient rule.

Mentor: That is not evidence. What exactly were u , v , u' , and v' ?

Student: Okay, I did not actually write it out. Let me do it now. $u = 2x + 5$, so $u' = 2$.
 $v = x^2 + 1$, so $v' = 2x$. Therefore,

$$y' = \frac{2(x^2 + 1) - (2x + 5)(2x)}{(x^2 + 1)^2}$$

Mentor: Good. That is the bluff-catching moment: I did not accept "I practiced" without details. Transfer check: why is the sign in the numerator minus, not plus?

Student: Because quotient rule is $u'v - uv'$. Plus would be product rule behavior.

Mentor: Passed.

Lesson 6: Chain Rule

Mentor: Video: Khan Academy, "Chain rule": <https://www.youtube.com/watch?v=0T0QrHO56qg>

Short summary: for a composite function, differentiate the outside and multiply by the derivative of the inside.

Practice task:

$$y = \sqrt{4x^3 + 2}$$

Name the inner function and differentiate.

Student: Inner function is $4x^3 + 2$. Since square root is power $1/2$,

$$y' = \frac{1}{2}(4x^3 + 2)^{-1/2}$$

Mentor: You found the outer derivative, but you forgot the inner derivative. What is $\frac{d}{dx}(4x^3 + 2)$?

Student: It is $12x^2$. Corrected:

$$y' = \frac{1}{2}(4x^3 + 2)^{-1/2} \cdot 12x^2 = \frac{6x^2}{\sqrt{4x^3 + 2}}$$

Mentor: Correct. Transfer check: differentiate $(3x^2 - 1)^5$.

Student: Outer derivative gives $5(3x^2 - 1)^4$. Inner derivative is $6x$, so

$$30x(3x^2 - 1)^4$$

Mentor: Passed.

Lesson 7: Implicit Differentiation

Mentor: Video: Khan Academy, "Implicit differentiation": <https://www.youtube.com/watch?v=5pPdtHF1Bhw>

Short summary: when y is not isolated, treat y as a function of x . Differentiating y^3 gives $3y^2 \frac{dy}{dx}$.

Practice task:

$$x^2y + y^3 = 10$$

Differentiate and solve for $\frac{dy}{dx}$.

Student: Derivative of x^2y is $2xy + x^2 \frac{dy}{dx}$. Derivative of y^3 is $3y^2$. So

$$2xy + x^2 \frac{dy}{dx} + 3y^2 = 0$$

Mentor: Product rule on x^2y was good, but y^3 needs chain rule because y depends on x . Try that term again.

Student: It should be $3y^2 \frac{dy}{dx}$. Then

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

Group terms:

$$\frac{dy}{dx}(x^2 + 3y^2) = -2xy$$

So

$$\frac{dy}{dx} = \frac{-2xy}{x^2 + 3y^2}$$

Mentor: Correct. Transfer check: what earlier rule did you use in x^2y ?

Student: Product rule, because x^2 and y are multiplied.

Mentor: Passed.

Lesson 8: Tangent Lines and Linear Approximation

Mentor: Video: Khan Academy, "Derivative as slope of a tangent line": <https://www.youtube.com/watch?v=ANyVpMS3HL4>

Short summary: a tangent line at $x = a$ uses the point and derivative:

$$y - f(a) = f'(a)(x - a)$$

It can approximate nearby function values.

Practice task: find the tangent line to $f(x) = x^3 - x$ at $x = 2$, then approximate $f(2.1)$.

Student: $f(2) = 8 - 2 = 6$. $f'(x) = 3x^2 - 1$, so $f'(2) = 11$. Tangent line:

$$y - 6 = 11(x - 2)$$

So $y = 11x - 16$. At $x = 2.1$, the approximation is **7.1**.

Mentor: Good. Now compare with the exact value.

Student: $f(2.1) = 2.1^3 - 2.1 = 9.261 - 2.1 = 7.161$. The tangent estimate is close and a little low.

Mentor: Why might it be low?

Student: Since $f''(x) = 6x$, near $x = 2$ the graph is concave up, so the tangent line lies below the curve.

Mentor: Passed. You used both derivative and interpretation.

Lesson 9: Optimization

Mentor: Video playlist: Khan Academy derivative applications / optimization:
<https://www.youtube.com/playlist?list=PLSQI0a2vh4HDHy9nIoEb-iQ9eQVYoXiHy>

Short summary: optimization means turning a word problem into a one-variable function, differentiating, setting the derivative equal to zero, and checking whether the result is a maximum or minimum.

Practice task: a rectangle has perimeter **40**. What dimensions maximize its area?

Student: $2l + 2w = 40$, so $l + w = 40$, and $w = 40 - l$. Then

$$A = l(40 - l) = 40l - l^2$$

Set $A' = 40 - 2l = 0$, so $l = 20$, $w = 20$.

Mentor: Check the perimeter equation. If $2l + 2w = 40$, what is $l + w$?

Student: I divided wrong. $2l + 2w = 40$ means $l + w = 20$, so $w = 20 - l$.

$$A = l(20 - l) = 20l - l^2$$

Then $A' = 20 - 2l$. Set $A' = 0$, so $l = 10$, $w = 10$. Maximum area is **100**.

Mentor: How do you know it is a maximum?

Student: $A'' = -2$, so the parabola opens downward and the critical point is a maximum.

Mentor: Passed.

Lesson 10: Related Rates

Mentor: Video: Khan Academy, "Related rates intro": <https://www.youtube.com/watch?v=kQF9pOqmS0U>

Short summary: related rates connect quantities that change over time. Write an equation, differentiate both sides with respect to time, substitute known values, then interpret sign and units.

Practice task: a 10 ft ladder leans against a wall. The bottom slides away at **1** ft/s. How fast is the top sliding down when the bottom is **6** ft from the wall?

Student: Let x be the bottom distance and y be the height. Then

$$x^2 + y^2 = 100$$

Differentiate:

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

At $x = 6$, $y = 8$, and $\frac{dx}{dt} = 1$:

$$12 + 16 \frac{dy}{dt} = 0$$

So $\frac{dy}{dt} = \frac{3}{4}$ ft/s.

Mentor: The magnitude is right, but the sign is wrong. If the bottom moves away, the top moves down, so $\frac{dy}{dt}$ must be negative. Solve the last step again.

Student: $16 \frac{dy}{dt} = -12$, so

$$\frac{dy}{dt} = -\frac{12}{16} = -\frac{3}{4}$$

The top slides down at **0.75** ft/s.

Mentor: Final transfer check: differentiate

$$y = \frac{(x^2 + 1)^3}{x - 1}$$

This combines quotient rule and chain rule.

Student: $u = (x^2 + 1)^3$, so $u' = 6x(x^2 + 1)^2$. $v = x - 1$, so $v' = 1$. Therefore,

$$y' = \frac{6x(x^2 + 1)^2(x - 1) - (x^2 + 1)^3}{(x - 1)^2}$$

Mentor: Course complete. You corrected errors, handled a bluff honestly, and transferred old rules to a new problem.